PARAM TULLY

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LinkedIn ⋄ GitHub ⋄ Portfolio

EDUCATION

University of British Columbia, BSc in Computer Science — Sept 2021 - May 2025
Relevant Coursework: Advanced Machine Learning, Applied Machine Learning, Artificial Intelligence, Linear Algebra, Statistics
Honours: Undergraduate Open Scholarship, President's Honour Roll, Dean's Honour List

EXPERIENCE

DevOps Engineer Intern — Invoke, Vancouver, BC — Sept 2023 – Dec 2023

- Cut mobile release cycle 6x (60 min to 10 min) by building a Fastlane and CircleCI pipeline for iOS builds, code signing, and App Store deployment
- Shipped production OAuth for Google, Facebook, and Apple using AWS Cognito across a React Native client and Node.js API
- Provisioned AWS infrastructure with Terraform and cut CI runtime by parallelizing tests and optimizing Docker layer caching

Technologies: TypeScript, React Native, AWS, Terraform, Docker, Fastlane, CircleCI

PROJECTS

AI Review Response App [asterism-app.vercel.app]

- Building an iOS and FastAPI product that drafts AI replies to business reviews in the owner's voice, with Google and Yelp OAuth clients built against a local stub while partner API access is pending
- Designed a RAG pipeline (pgvector, sentence transformer embeddings) that retrieves up to 3 similar past replies as few shot examples once a business has at least 8 posted replies and match similarity clears a set threshold
- Defended the draft pipeline against prompt injection by separating system instructions from customer review text, verified with adversarial prompt structure tests

Technologies: Python, FastAPI, PostgreSQL, pgvector, sentence-transformers, Swift, Supabase

Graph Neural Networks from Scratch

- Implemented a graph convolutional network and a graph transformer from scratch in PyTorch, including custom message passing layers and multi head self attention with Laplacian positional encodings
- Benchmarked both against PyTorch Geometric reference layers on Peptides-struct; my GCN landed within 5% of the reference (0.95 vs 0.91 test MSE), while my transformer fit training data harder yet tested worse, isolating a generalization gap

Technologies: Python, PyTorch, PyTorch Geometric

Credit Default Prediction

- Compared logistic regression, SVM, random forest, and XGBoost on 30,000 credit clients; the best model reached 81% test accuracy against a 78% majority class baseline, so I scored on precision, recall, and F1 instead

Technologies: Python, pandas, scikit-learn, XGBoost

SKILLS

Languages: Python, SQL, TypeScript, Bash
ML & AI: PyTorch, RAG, embeddings, pgvector, sentence-transformers, attention mechanisms, scikit-learn, XGBoost
Stack: FastAPI, PostgreSQL, pandas, NumPy, AWS Lambda, GitHub Actions